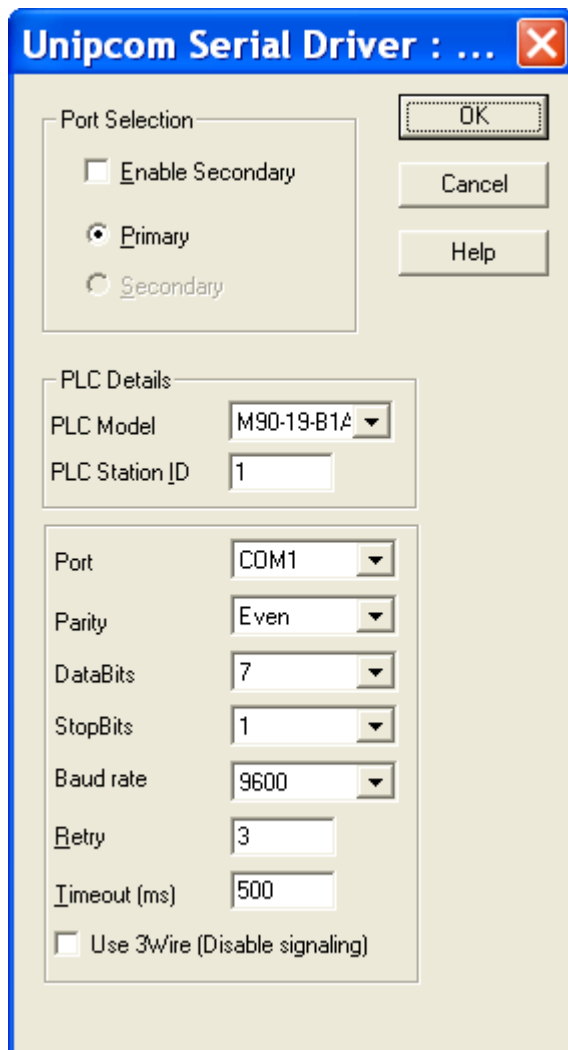


UNITRONICS PCOM_NET protocol driver

Overview

Adroit supports communication with the following Unitronics Devices:
M90-19-B1A or UNRESTRICTED for the other modifiable models.
This driver has be tested with M-90 and a M-210.



The image shows a Windows-style dialog box titled "Unipcom Serial Driver : ...". It contains three main sections: "Port Selection", "PLC Details", and a section for serial communication parameters. The "Port Selection" section has three radio buttons: "Enable Secondary" (unchecked), "Primary" (checked), and "Secondary" (disabled). The "PLC Details" section has a "PLC Model" dropdown menu set to "M90-19-B1A" and a "PLC Station ID" text box containing "1". The serial parameters section includes dropdown menus for "Port" (COM1), "Parity" (Even), "DataBits" (7), "StopBits" (1), and "Baud rate" (9600). It also has text boxes for "Retry" (3) and "Timeout (ms)" (500). At the bottom, there is a checkbox for "Use 3Wire (Disable signaling)" which is unchecked. On the right side of the dialog, there are three buttons: "OK", "Cancel", and "Help".

Section	Parameter	Value
Port Selection	Enable Secondary	☐
	Primary	☒
	Secondary	☐
PLC Details	PLC Model	M90-19-B1A
	PLC Station ID	1
Serial Parameters	Port	COM1
	Parity	Even
	DataBits	7
	StopBits	1
	Baud rate	9600
	Retry	3
	Timeout (ms)	500
	Use 3Wire (Disable signaling)	☐

PLC Details

PLC Model – Select ‘M-90’ for standard M-90
Select ‘Unrestricted’ if the device is extended unrestricted provides for protocol maximums (not Device maximums).
PLC Station ID – Network address if networked.

Local Details

Port – The available serial ports on your machine.

Parity – default : even

DataBits – default : 7

StopBits – default : 1

Baud rate – default : 9600, (9600,19200,38400,57600)

Retry - The number of times that Adroit will try to gain communication with a PLC before failing a transaction.

Timeout - The time that Adroit will wait for a valid response from a PLC before failing a transaction (Read or Write).

Use 3Wire (Disable signaling) – (Default is off i.e. signalling enabled)
Check this if you want to use 3 wire only.

Note for M-2xx series users:

The M-2xx series must have a function block loaded via Unitronics programming software !

Eg PCOM_NET , PCOM_NS

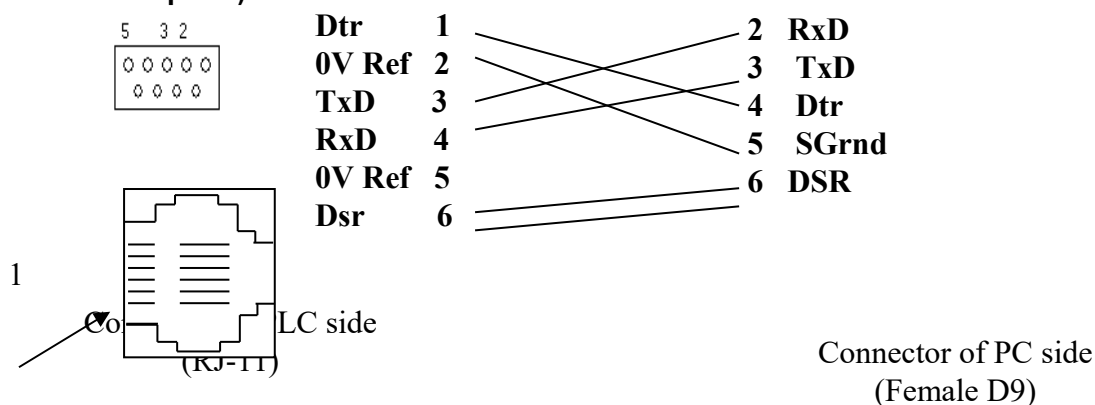
where various settings are made available eg Addressing, Rs232 settings not normally available from the faceplate!

NB – if retries persist continually : try lowering DTR delay settings in the function block!

NB – Make sure if you have hardware handshaking enabled use the correct rs232c cable (not 3 wire)

Wiring

Cable connection for RS232C communication (Connection between PLC and Computer)



Supported addresses

M90 PLC data address types and maximum ranges supported by the Adroit M90 Protocol driver.

Device	Min Range	Max Range	PLC Data Type	BOOLEAN	INTEGER	REAL	Read	Write
DI Area	0	*1	Bit	DI:0	-	-	Y	N
DO Area	0	*2	Bit	DO:0	-	-	Y	Y
MI Area	0	*3	Unsigned 32bit Signed 32bit	- -	MI:0 W MI:0 I	MI:0 W MI:0 I	Y Y	Y Y
MB Area	0	*4	Bit	MB:0	-	-	Y	Y
SI Area	0	*5	Unsigned 32bit Signed 32bit	-	SI:0 W SI:0 I	SI:0 W SI:0 I	Y Y	N N

M-90 model can be extended , hence

- *1 -not larger than 255 (M90 Device limit), 2048(Protocol limit)
- *2 -not larger than 255 (M90 Device limit), 2048(Protocol limit)
- *3 -not larger than 255 (M90 Device limit), 65535(Protocol limit)
- *4 -not larger than 255 (M90 Device limit), 65535(Protocol limit)
- *5 -not larger than 255 (M90 Device limit), 65535(Protocol limit)

Size specifier (eg W) for Integers and Reals are not used but the default (W) is specified , this is to cater for possible future changes

Message protocol

Protocol overview

Read request:

STX	Station	Function	Start Addr	Length	Bcc	ETX
/	01	DI	00,0	04	XX	/r

Read response:

STX	ACK	Station	Function	Start Addr	Length	Data bytes	Bcc	EOT
/	A	01	DI	00,0	04	00 00 00 00	XX	/r

Write request:

STX	Station	Function	Start Addr	Length	Data	Bcc	ETX
/	01	DO	00,0	01	01	XX	/r

Write response:

STX	ACK	Station	Function	Start Addr	Length	Bcc	ETX
/	A	01	DI	00,0	01	XX	/r

The following transactions are supported:

- 1.Read (RE) Digital Input
- 2.Read (RA) Digital Output
- 3.Write (SA) Digital Output
- 4.Read (RW) Internal Integer
- 5.Write (SW) Internal Integer
- 6.Read (RB) Internal Bit
- 7.Write (SB) Internal Bit
- 8.Read (GF) System Integer

BCC:

The Bcc is obtained by adding every byte except STX,ETX and Bcc and Ack, truncated to a Byte , where each nibble is converted to its corresponding Ascii value.
Eg 40h+41h+43h = 124h =Truncated> 24h =Converted (3234h or '24'Ascii)

Revision information

Date	Changes
29/05/2000	Revision 0 – 1 st release.
01/09/2006	Revision 5 – Added a 3wire disable signalling checkbox.